
Sys One: Fast, Selective Tool-Call Routing in a Single Forward Pass

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Abstract

When an AI assistant must decide *which tool to call*, the default approach is to prompt a large language model to choose. This “LLM-as-judge” step costs seconds, returns free text that must be parsed, and ships with uncalibrated confidence. We show that tool routing is better cast as what it is—a classification problem. A 0.6B-parameter model with a sequence-classification head, fine-tuned with LoRA on open-source function-calling data, routes among 25 tool classes in a *single forward pass*: 99.3% accuracy and 0.989 macro-F1 on a held-out test set, ~ 30 ms per decision on a GTX 1080 Ti, emitting per-class probabilities as typed output.

We frame routing as *selective classification*: classify $\rightarrow$ calibrate $\rightarrow$ accept or abstain $\rightarrow$ escalate. Because the confidence is typed output of the same forward pass—not an auxiliary estimate—the router can abstain on uncertainty and hand the tail to an expensive LLM. At a 0.9 confidence threshold the router accepts 98.1% of traffic at 99.88% selective accuracy and escalates 1.9%; 10 of its 12 test errors fall below the threshold. Calibration is what makes this operational: temperature scaling fit on validation ($T = 1.79$) corrects genuine overconfidence and improves test NLL ($0.0310 \rightarrow 0.0261$) and Brier score ($0.0131 \rightarrow 0.0120$), while ECE ticks up slightly ($0.0067 \rightarrow 0.0077$)—reported honestly rather than hidden. A companion experiment on easier data collapsed to $T = 1.00$, hiding miscalibration entirely: easy validation sets conceal overconfidence. We release code, curated datasets, and evaluation artifacts.

1 Introduction

Modern AI assistants act through tools: calculators, search APIs, calendars, messengers, and domain backends [Qin et al., 2023, Schick et al., 2023]. Before any tool executes, the system must solve a routing problem: given the user’s utterance, *which tool, if any, should handle it?* The dominant engineering pattern delegates this decision to a large language model prompted to select a tool [Yao et al., 2023, Zheng et al., 2023]. This works, but it is expensive in the exact dimension agents cannot afford: an extra LLM call adds 2–30 seconds of latency to every agentic step, returns free-form text that must be parsed into a structured call, and produces confidence estimates that are poorly calibrated [Kadavath et al., 2022, Guo et al., 2017].

We argue the routing decision should not be generated at all. It should be *classified*. A small encoder-style model with a linear head maps an utterance to a distribution over a fixed tool catalog in one forward pass, yielding typed output (tool, probabilities, confidence) with latency measured in milliseconds rather than seconds. Small specialized models have already proven surprisingly capable at agentic subtasks—Salesforce’s 1B-parameter xLAM “tiny giant” beats much larger models on function calling [Zhang et al., 2024]—but most work still frames tool selection as generation. We take the classification view seriously and ask: how far does a 0.6B model go, and can its confidence be trusted?

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The production architecture this enables is *classify* $\rightarrow$ *calibrate* $\rightarrow$ *accept or abstain* $\rightarrow$ *escalate*: a small classifier handles high-confidence routing locally in ~ 30 ms and escalates uncertain cases to the expensive LLM. This is the paper’s center: tool routing as *selective classification* [Geifman and El-Yaniv, 2017]. Classification is what makes the architecture deployable—the confidence that drives abstention is typed output of the same forward pass, not an auxiliary estimate bolted on afterward. Calibration, in this framing, is not the finding; it is what makes the finding operational.

Our contributions:

- A data-curation pipeline that turns the Glaive v2 function-calling corpus [glaiveai, 2024] into a clean 25-way routing benchmark: 949 raw tool names normalized to 24 canonical tools plus a `no_tool` class, exact-duplicate removal (without which train/test leak identical strings), and majority resolution of label conflicts (10,936 examples).
- A LoRA [Hu et al., 2022] training recipe for Qwen3-0.6B [Qwen Team, 2025] that is stable on older (Pascal) GPUs: fp32 frozen backbone and adapters with fp16 autocast compute, class-weighted loss for a long-tailed label distribution.
- A selective-routing analysis with risk–coverage curves: at a 0.9 confidence threshold the router accepts 98.1% of test traffic at 99.88% selective accuracy, escalating 1.9% to a larger model and catching 10 of its 12 errors. Calibration is presented as the enabler: temperature scaling [Guo et al., 2017] fit on validation gives $T = 1.79$ —genuine, measurable overconfidence—and improves held-out NLL and Brier score. A companion binary experiment on easy template data gave $T = 1.00$, i.e., *no detectable miscalibration*, which was wrong: the validation set was simply too easy to expose it. Calibration measured on unrepresentative data is not calibration.
- Latency numbers on commodity hardware: mean 29.8 ms per decision including tokenization on a single GTX 1080 Ti ($\sim 100\times$ faster than LLM-as-judge), 175–190 decisions/sec batched.

2 Related Work

Tool use in language models. ReAct [Yao et al., 2023] interleaves reasoning and acting; Toolformer [Schick et al., 2023] teaches models to invoke APIs via self-supervised objectives; Gorilla [Patil et al., 2023] connects LLMs to massive API stores with retrieval. These frame tool selection as generation or retrieval. We instead treat routing as classification. For agents operating over a fixed or slowly changing tool catalog, routing admits a simpler formulation: discriminative classification over the catalog.

Small and specialized models for agents. The xLAM family [Zhang et al., 2024] demonstrates that compact models trained on high-quality synthetic agent data (APIGen [Liu et al., 2024]) can match or beat frontier models on function-calling benchmarks such as BFCL [Berkeley Artificial Intelligence Research, 2024]. Our work shares the small-model thesis but differs in mechanism: one discriminative forward pass instead of autoregressive generation, which removes parsing entirely and makes latency predictable.

Calibration. Guo et al. [Guo et al., 2017] showed modern neural networks are poorly calibrated and that temperature scaling is a simple, effective remedy; Naeini et al. [Naeini et al., 2015] formalized expected calibration error; the Brier score [Brier, 1950] remains a standard proper scoring rule. Desai and Durrett [Desai and Durrett, 2020] found pre-trained transformers miscalibrated out of the box, while Kadavath et al. [Kadavath et al., 2022] showed language models often “know what they know” when asked properly. LLM-as-judge pipelines [Zheng et al., 2023] rarely calibrate at all. Our contribution here is empirical and cautionary: we exhibit a concrete case where temperature scaling reports perfect calibration ($T = 1.00$) on easy validation data while the same procedure on representative data reveals substantial overconfidence ($T = 1.79$).

Parameter-efficient adaptation. We build on LoRA [Hu et al., 2022] and the sequence-classification fine-tuning paradigm established by BERT [Devlin et al., 2019], applied here to the Qwen3 family [Qwen Team, 2025].

3 Method

3.1 Task formulation

Let x be a user utterance and $\mathcal{Y} = \{1, \dots, 25\}$ a fixed set of routing classes (24 canonical tools plus `no_tool`). A router is a classifier $p_\theta(y | x)$. The routing decision is $\hat{y} = \arg \max_y p_\theta(y | x)$ with confidence $c = \max_y p_\theta(y | x)$; downstream systems can threshold c for selective abstention [Geifman and El-Yaniv, 2017]. The full distribution is emitted as typed output, so callers get uncertainty, not just a label.

3.2 Data curation

We build on `glaiveai/glaive-function-calling-v2` [glaiveai, 2024] (Hugging Face, Apache-2.0, 112,960 rows), chosen over alternatives such as `Salesforce/xlam-function-calling-60k` because its v2 split contains natural no-call turns (the user asks for something the assistant cannot do $\rightarrow$ refusal), which become the `no_tool` class.

Extraction. Each chat is split into turns. Every assistant turn containing a `<functioncall>` yields the pair (immediately preceding user utterance, tool name); rows with no function call yield (last user utterance, `no_tool`). Multi-turn chats can yield several examples.

Normalization. The raw data contains 949 distinct tool names with a heavy synonym tail (`get_news` / `get_news_headlines` / `search_news`; `generate_password` / `generate_random_password`; `create_event` / `schedule_meeting`; ...). A synonym map merges these into the top 24 canonical tools plus `no_tool`: 25 classes. The full list is in Appendix A.

Deduplication. Glaive v2 is heavily templated: e.g., `calculate_bmi` appears 3,103 times but only 171 utterances are unique. Exact (text, label) dedup is load-bearing—without it, train and test share identical strings and the benchmark is meaningless. 128 texts map to multiple tools (generic clarifications such as “what do you mean?”); we keep the majority label.

The final dataset has **10,936 examples**, stratified 70/15/15 into train (7,655), validation (1,640), and test (1,641). Per-class counts range from 37 to 700 tool examples; `no_tool` has 2,000.

3.3 Model and training

We use Qwen3-0.6B [Qwen Team, 2025] in non-thinking mode (no chain-of-thought) with a sequence-classification head:

$$p_\theta(y | x) = \text{softmax}(Wh_\phi(x) + b), \quad (1)$$

where $h_\phi(x)$ is the pooled final hidden state. We fine-tune with LoRA [Hu et al., 2022], rank 16 on all attention and MLP projections, with the score head fully trained: $\sim 10.1\text{M}$ trainable parameters (1.67%).

Stability recipe. On Pascal-era GPUs, full-fp16 training diverged. We keep the frozen backbone and the LoRA adapters in fp32 and use fp16 only via autocast for compute. This recipe trained without a single divergence.

Optimization. 3 epochs, batch size 16, learning rate 2×10^{-4} with cosine decay and 100 warmup steps, early stopping on validation macro-F1. The label distribution is long-tailed (`generate_random_number`: 37 train examples vs. `no_tool`: 1,400), so we use a class-weighted cross-entropy, $\mathcal{L} = -\sum_i w_{y_i} \log p_\theta(y_i | x_i)$ with $w_c \propto 1/n_c$.

3.4 Calibration

We apply temperature scaling [Guo et al., 2017], $p_T(y | x) = \text{softmax}(z/T)$, fitting the single scalar T by minimizing NLL on the *validation* split only—never on test. This is the correct protocol and, as Section 4.3 shows, the choice of validation data determines whether calibration means anything.

Table 1: Leaderboard: routing quality and latency by dataset and task. Every row is a setting with a train/test split on which a model was fine-tuned; the generative column is the stronger 3-shot baseline using the same frozen Qwen3-0.6B in non-thinking mode (plain-text prompts, greedy decoding, ≤ 16 tokens, no confidence output). Parse failures count as wrong (28.1% zero-shot, 15.2% 3-shot on the curated set). Latency is mean per-example, fp16, including tokenization: GTX 1080 Ti for the curated row; the Glaive-100 row[†] was measured on an RTX 3090 (different GPU—compare with care); the Glaive-200 row[‡] was measured on another RTX 3090 host whose 2014-era CPU makes single-sample latency launch-overhead bound—batch throughput is the informative figure (see text). Macro-F1 is over the classes present in each split, with parse failures as errors. Datasets without an official train/test split are excluded from the leaderboard by design—BFCL is flagged for future LoRA/transfer-learning work. Temperature is fit on each setting’s validation split only ($T = 1.7912$ curated, $T = 1.4063$ Glaive-100, $T = 1.3922$ Glaive-200). The generative baselines for the Glaive-100 and Glaive-200 settings are pending (—) and will complete the rows when run.

Dataset + task	Accuracy		Macro-F1		Latency (ms)	
	Gen.	Sys One	Gen.	Sys One	Gen.	Sys One
Curated, tool routing ($n=1,641$)	0.512	0.993	0.523	0.989	500.6	29.8
Glaive-100, tool routing ($n=2,557$)	—	0.960	—	0.880	—	22.2[†]
Glaive-200, tool routing ($n=2,805$)	—	0.939	—	0.845	—	55.0[‡]

4 Experiments

4.1 Setup

All experiments run on a single NVIDIA GTX 1080 Ti, except the two Glaive scale runs, which were trained and evaluated on rented RTX 3090s while the local GPU was unavailable. The base model is Qwen3-0.6B in non-thinking mode, frozen, for every experiment—both the classifier and the generative baseline. Metrics: accuracy, macro-F1, NLL, Brier score [Brier, 1950], and ECE with 15 bins [Naeini et al., 2015]. Latency is measured including tokenization.

4.2 Main results

Table 1 is the paper in one table: the classifier reaches **99.3% accuracy** at **29.8 ms** per decision, while the matched generative baseline—same base model, same 25 tools, same 1,641 queries—manages 51.2% (3-shot) at 500.6 ms. That is a $\sim 17\times$ **latency win** and a **+48pp accuracy gain** at once; the usual latency–quality tradeoff never appears because the baseline pays for free-text generation twice: in milliseconds, and in the 15–28% of outputs that fail to parse into a valid tool call. Temperature scaling leaves accuracy and F1 unchanged (it is monotonic in the logits) while improving the proper scoring rules that measure uncertainty quality: NLL $0.0310 \rightarrow 0.0261$, Brier $0.0131 \rightarrow 0.0120$. ECE ticks *up* slightly ($0.0067 \rightarrow 0.0077$); with ECE already near zero this is noise, but it is an honest reminder that temperature scaling improves likelihood, not necessarily every calibration statistic.

Scaling to a larger catalog preserves the picture. On the 101-class Glaive-100 setting (100 tools + no_tool, $n = 2,557$), the same frozen-Qwen3-0.6B classifier reaches **96.0% accuracy / 0.880 macro-F1** at **22.2 ms** per decision. The accuracy drop from the 25-class setting concentrates in the thin tail—26 of 101 classes have fewer than 40 post-deduplication training examples, with test supports as small as 1–3—and in near-synonym confusions (get_recipe $\rightarrow$ search_recipes, get_news $\rightarrow$ get_news_articles). Selective prediction at confidence ≥ 0.9 covers 88.7% of queries at 99.0% accuracy, rejecting 79 of the 101 test errors.

At 179 classes the trend continues. The Glaive-200 setting (178 tools + no_tool, $n = 2,805$) reaches **93.9% accuracy / 0.845 macro-F1**. The drop again concentrates in the thin tail—add_task, analyze_customer_reviews, and create_account all score F1 = 0 on tiny supports—and in near-synonym pairs (search_movies $\rightarrow$ get_movie_recommendations, get_recipe $\rightarrow$ search_recipes, get_news_articles $\rightarrow$ get_news, 6 confusions each). Selective prediction at confidence ≥ 0.9 holds 85.6% coverage at 98.1% accuracy, rejecting 125 of the 171 test errors. Per-decision latency on the rented 3090 host measured 55.0 ms, but that figure is launch-

Table 2: Retrieval baselines vs. Sys One. BM25 and TF-IDF over per-tool descriptions (names only for the curated catalog); `no_tool` via a score threshold tuned on each validation split for macro-F1. No training, CPU-only.

Setting	BM25		TF-IDF		Sys One	
	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1
Curated ($n=1,641$)	0.604	0.602	0.605	0.611	0.993	0.989
Glaive-100 ($n=2,557$)	0.465	0.435	0.492	0.473	0.960	0.880
Glaive-200 ($n=2,805$)	0.410	0.380	0.447	0.425	0.939	0.845

overhead bound by the host’s 2014-era CPU and is not comparable across rows; batch throughput—257 queries/s at batch 16, 1,055/s at batch 64—is the informative number.

A natural objection is that routing needs no trained model at all—just retrieve the most similar tool description. Table 2 tests it: BM25 and TF-IDF over per-tool descriptions (names only for the curated catalog, which ships none), with `no_tool` predicted when the top score falls below a threshold tuned on each validation split for macro-F1. At 25 tools the untrained baselines are respectable—0.605 accuracy, ahead of the 3-shot generative baseline’s 0.512—but they collapse as the catalog grows (0.605 $\rightarrow$ 0.447) while the trained head holds (0.993 $\rightarrow$ 0.939). Near-synonym tools (`get_recipe` vs. `search_recipes`) collide lexically; the classifier learns the fine distinctions that similarity cannot see.

Datasets without an official train/test split are excluded from the leaderboard by design. BFCL [Berkeley Artificial Intelligence Research, 2024] ships no train/test split, so a frozen transfer probe over 68 manually mapped functions (69 single-function examples) cannot be a leaderboard entry—there is no split to fine-tune against. We flag BFCL, and any similar split-less dataset, for future LoRA/transfer-learning work, where the protocol would be to construct a split and fine-tune rather than probe frozen transfer.

4.3 Calibration: what easy validation hides

The $T = 1.79$ fit is the interesting number in this paper. It says the model, like most modern networks [Guo et al., 2017], is genuinely overconfident—and that a one-parameter fix, fit on representative data, measurably improves its probabilities.

Contrast this with a companion binary experiment (prompt-injection detection on template-synthesized data) where the same procedure returned $T = 1.00$: *no detectable miscalibration*. That reading was wrong, not lucky. The validation set was so easy that the model’s confidence was never stressed, so temperature scaling had nothing to find. Two lessons follow. First, calibration is a property of the (model, data-distribution) pair, not of the model alone; a T fit on unrepresentative data certifies nothing. Second—and more practically—the same illusion applies to accuracy: benchmarks built from templated data flatter both correctness and confidence. Our dedup (Section 3.2) removes the leakage that makes this worse, but paraphrase-level template artifacts remain (Section 5).

4.4 Selective routing: the architecture this enables

Classification earns its keep when the router is allowed to say “I don’t know.” Because confidence is typed output of the same forward pass, a threshold on $c = \max_y p_T(y | x)$ implements selective classification [Geifman and El-Yaniv, 2017]: accept high-confidence decisions locally, escalate the rest to the expensive LLM. Table 3 gives the risk–coverage curve on the test set.

At a 0.9 threshold the router accepts 98.1% of traffic at 99.88% selective accuracy, escalating 1.9%. The errors are catchable: 10 of the 12 test errors fall below 0.9 confidence (mean error confidence 0.69 vs. 0.99 for correct decisions), while only 1.3% of correct decisions sit below the threshold—so abstention buys nearly all of the error reduction at negligible escalation cost. The `no_tool` class needs no special handling: it is predicted with 0.997 precision and 0.993 recall, i.e., the router decides that *no* tool applies confidently and correctly; the abstained tail is dominated by genuinely ambiguous tool–tool boundaries.

Table 3: Risk–coverage on the test set ($n = 1,641$), calibrated confidences ($T = 1.791$). Coverage is set by thresholding confidence.

Coverage	Threshold	Selective acc.	Errors	Escalated
100%	—	0.9927	12	0
99%	0.69	0.9969	5	16
98%	0.91	0.9988	2	33
95%	0.99	0.9994	1	82
90%	0.996	1.0000	0	164

Table 4: Inference latency, fp16 including tokenization, single GTX 1080 Ti.

Setting	
Single sample, mean	29.8 ms
Single sample, p50 / p99	29.3 / 47.9 ms
Throughput, batch 16	175 samples/sec
Throughput, batch 64	190 samples/sec

The practical claim is therefore stronger than “a small model is faster”: a small classifier handles high-confidence routing locally in ~ 30 ms and escalates uncertain cases to the expensive LLM. The escalation path also bounds the damage of Section 5: where the ontology is fuzzy, the router abstains instead of guessing.

4.5 Latency

Per-decision latency is ~ 30 ms end to end (Table 4), $\sim 17\times$ faster than the matched generative baseline on identical hardware (Table 1)—and two orders of magnitude faster than a hosted LLM-as-judge call (typically 2–30 s)—with typed output and no parsing step. Batching lifts throughput to ~ 190 decisions/sec on decade-old hardware.

Latency wins compound into cost wins. At the \$0.1422/hr rental rate of the RTX 3090 host, the 179-class router’s batch-64 throughput (1,055 queries/s) prices routing at roughly **\$0.04 per million queries**; at single-query latency the same hardware is \$2.17 per million. The 3-shot generative baseline (500.6 ms per query on the 25-tool catalog) costs \$19.77 per million at the same hourly rate—about $9\times$ the single-query cost and $530\times$ the batched cost, on identical 0.6B parameters. The gap widens structurally with catalog size: the generative prompt must carry every tool definition, so its token count grows linearly with N ($\sim 1,550$ tokens at 100 tools, $\sim 2,744$ at 178, by a chars/4 estimate over the measured descriptions), while the classifier’s per-query cost is $O(1)$ in N .

4.6 Error analysis

The weakest classes are the small and near-duplicate ones: `calculate_mortgage_payment` (F1 0.933, $n=32$), `get_movie_details` (0.964), `translate_text` (0.966), `calculate_loan_payment` (0.967), `get_definition` (0.970). The top confusions are *dataset-level tool duplicates*, not model failures: `calculate_mortgage_payment` $\rightarrow$ `calculate_loan_payment` ($\times 4$) and `search_movies` $\rightarrow$ `get_movie_details` ($\times 2$) are near-synonyms in the source data; a production router would merge them into one canonical tool. The genuinely interesting boundary cases are the three `no_tool` confusions (one `translate_text` $\rightarrow$ `no_tool`, two in the reverse direction): deciding that *no* tool applies is the hardest part of routing, and the part most worth calibrating.

The deeper point is about the catalog, not the model: *router quality is upper-bounded by tool ontology quality*. When two classes are near-synonyms in the data—mortgage vs. loan payment, movie search vs. movie details—no classifier can separate them reliably, because there is nothing to separate. The fix is in the ontology (merge the duplicates), not in more parameters. This is a systems observation, not a modeling one, and it is good news: ontology cleanup is cheaper than model iteration.

5 Limitations

1. **Templated source data.** Glaive v2 utterances are synthetic and templated. Dedup removed exact repeats, but paraphrase-level template artifacts remain: the model partly learns “which template family is this” rather than deep intent. Real assistant traffic with human-verified labels would be strictly better.
2. **Utterance-only input.** The router sees only the final user utterance—no conversation history, no tool descriptions. Follow-ups like “what about France?” are genuinely ambiguous without context.
3. **Thin tail classes.** `generate_random_number` (37 train), `translate_text` (69), and `get_definition` (74) have few unique utterances; expect weak F1 there.
4. **One decision per forward pass.** The router makes a single routing decision per pass; it does not score multiple parallel tool calls jointly.
5. **No teacher labeling.** Labels come from the dataset’s own function-call annotations, taken at face value, with no human review.

6 Conclusion

Tool-call routing does not need generation. A 0.6B-parameter classifier, fine-tuned with LoRA on curated open-source data, routes among 25 tool classes at 99.3% accuracy in ~ 30 ms per decision, emitting typed probabilities with calibrated confidence. The durable lesson is architectural: routing is *selective classification*—classify, calibrate, accept or abstain, escalate. A small classifier handles high-confidence routing locally and hands its uncertain tail to the expensive LLM; at a 0.9 threshold, 98.1% of traffic is accepted at 99.88% selective accuracy while 1.9% is escalated. Calibration is what makes this operational, and it must be measured against the distribution served: temperature scaling fit on easy validation data reported perfect calibration ($T = 1.00$) that was an artifact of the data, while representative data exposed real overconfidence ($T = 1.79$) and fixed it. Finally, router quality is upper-bounded by tool ontology quality—several “errors” are duplicate tools in the catalog, and the fix is to merge them, not to scale the model.

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A Canonical tool classes

The 25 routing classes (24 canonical tools + no_tool): `analyze_sentiment`, `calculate_age`, `calculate_area`, `calculate_bmi`, `calculate_discount`, `calculate_distance`, `calculate_loan_payment`, `calculate_mortgage_payment`, `calculate_tip`, `convert_currency`, `create_calendar_event`, `create_todo`, `generate_password`, `generate_qr_code`, `generate_random_number`, `get_definition`, `get_movie_details`, `get_news`, `get_stock_price`, `no_tool`, `search_books`, `search_movies`, `search_recipes`, `send_email`, `translate_text`. The exact label map is `label_map.json` in the repository.

B Reproducibility

Code, datasets (`data/data_train.jsonl` / `data/data_val.jsonl` / `data/data_test.jsonl`), label map, fitted temperature, and the evaluation report are in the public repository. `requirements.txt` pins the exact dependency versions. Reference run: `python scripts/make_data.py` → `python scripts/train.py` (~20min on a GTX 1080 Ti) → `python eval.py`. Trained weights (~2.7 GB) are not committed; `train.py` reproduces them.